Quiz 1 – Detailed Solutions

Problem 1

Let $(Y_1, Y_2, Y_3)^\top \sim N_3\left((\mu_1, \mu_2, \mu_3)^\top, \Sigma\right)$ *where* $\Sigma = (1-\rho)I_3 + \rho\mathbf{1}\mathbf{1}^\top$.

Part (a)

Define $\mathcal{Z} = (Y_1 - Y_2, Y_2 - Y_3, Y_1 + Y_2 + Y_3)^\top$. *Find the distribution of* $\mathcal{Z}$.

Theory: For a linear transformation of a multivariate normal vector, say $Z = AY$, we have

$$Z \sim N_p(A\mu,\ A\Sigma A^\top).$$

We will compute the mean vector and covariance matrix explicitly.

Solution: Let

$$A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{pmatrix}, \quad \mu = (\mu_1, \mu_2, \mu_3)^\top.$$

Then $\mathcal{Z} = AY$, so

$$E[\mathcal{Z}] = A\mu = \begin{pmatrix} \mu_1 - \mu_2 \\ \mu_2 - \mu_3 \\ \mu_1 + \mu_2 + \mu_3 \end{pmatrix}.$$

Now compute the covariance:

$$\mathrm{Var}(\mathcal{Z}) = A\Sigma A^\top.$$

Since $\Sigma = (1-\rho)I_3 + \rho\mathbf{1}\mathbf{1}^\top$, we have

$$A\Sigma A^\top = (1-\rho)AA^\top + \rho(A\mathbf{1})(A\mathbf{1})^\top.$$

First,

$$A\mathbf{1} = \begin{pmatrix} 1-1+0 \\ 0+1-1 \\ 1+1+1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}.$$

Next,

$$AA^\top = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Therefore,

$$\mathrm{Var}(\mathcal{Z}) = (1-\rho)\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} + \rho\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 9 \end{pmatrix} = \begin{pmatrix} 2(1-\rho) & -(1-\rho) & 0 \\ -(1-\rho) & 2(1-\rho) & 0 \\ 0 & 0 & 3(1-\rho)+9\rho \end{pmatrix}.$$

Simplify the (3,3) entry: $3(1-\rho)+9\rho = 3+6\rho$.

Thus

$$\boxed{\mathcal{Z} \sim N_3\left(\begin{pmatrix}\mu_1-\mu_2\\ \mu_2-\mu_3\\ \mu_1+\mu_2+\mu_3\end{pmatrix}, \begin{pmatrix}2(1-\rho) & -(1-\rho) & 0\\ -(1-\rho) & 2(1-\rho) & 0\\ 0 & 0 & 3+6\rho\end{pmatrix}\right).}$$

Part (b)

Hence, or otherwise, find the conditional distribution of $(Y_1-Y_2, Y_2-Y_3)^\top$ given $\bar{Y}$.

Theory: If (Z_1, Z_2, Z_3) are jointly normal, then the conditional distribution of (Z_1, Z_2) given $Z_3 = z_3$ is normal with mean

$$\mu_{12} + \Sigma_{12}\Sigma_{22}^{-1}(z_3-\mu_3)$$

and covariance

$$\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21},$$

where the covariance matrix of (Z_1, Z_2, Z_3) is partitioned as

$$\begin{pmatrix}\Sigma_{11} & \Sigma_{12}\\ \Sigma_{21} & \Sigma_{22}\end{pmatrix}.$$

Solution: From part (a), let $Z_1 = Y_1 - Y_2$, $Z_2 = Y_2 - Y_3$, $Z_3 = Y_1 + Y_2 + Y_3 = 3\bar{Y}$. We see that the covariance between (Z_1, Z_2) and Z_3 is zero, i.e. $\Sigma_{12} = 0$ and $\Sigma_{21} = 0$. Hence they are independent (jointly normal and uncorrelated). Therefore,

$$(Z_1, Z_2) \perp Z_3,$$

so the conditional distribution of (Z_1, Z_2) given Z_3 (or given $\bar{Y}$) is the same as its unconditional distribution.

Thus

$$\boxed{(Y_1-Y_2,\ Y_2-Y_3)^\top \mid \bar{Y} \sim N_2\left(\begin{pmatrix}\mu_1-\mu_2\\ \mu_2-\mu_3\end{pmatrix}, \begin{pmatrix}2(1-\rho) & -(1-\rho)\\ -(1-\rho) & 2(1-\rho)\end{pmatrix}\right).}$$

Part (c)

Define $\underline{W} = \underline{Y} + \underline{q}$ where $\underline{q} \in \mathbb{R}^3$. Find the conditional distribution of $\underline{W}$ given $\bar{Y}$.

Theory: For a multivariate normal vector Y with mean μ and covariance Σ, the conditional distribution of Y given $\bar{Y} = \frac{1}{3}\mathbf{1}^\top Y$ is again normal. Its mean and variance can be derived using the joint normality of $(Y, \bar{Y})$.

Solution: We first find the conditional distribution of Y given $\bar{Y}$. Let $\bar{Y} = \frac{1}{3}(Y_1 + Y_2 + Y_3)$. Then

$$E[\bar{Y}] = \frac{\mu_1+\mu_2+\mu_3}{3} =: \bar{\mu}.$$

Compute the variance of $\bar{Y}$:

$$\mathrm{Var}(\bar{Y}) = \frac{1}{9}\mathbf{1}^\top\Sigma\mathbf{1}.$$

Now $\Sigma\mathbf{1} = (1-\rho)\mathbf{1} + \rho\mathbf{1}(\mathbf{1}^\top\mathbf{1}) = (1-\rho)\mathbf{1} + 3\rho\mathbf{1} = (1+2\rho)\mathbf{1}$, so $\mathbf{1}^\top\Sigma\mathbf{1} = \mathbf{1}^\top(1+2\rho)\mathbf{1} = 3(1+2\rho)$. Hence

$$\mathrm{Var}(\bar{Y}) = \frac{3(1+2\rho)}{9} = \frac{1+2\rho}{3}.$$

The covariance between Y and $\bar{Y}$ is

$$\mathrm{Cov}(Y,\bar{Y}) = \frac{1}{3}\Sigma\mathbf{1} = \frac{1+2\rho}{3}\mathbf{1}.$$

Thus, using the conditional normal formula,

$$E[Y \mid \bar{Y}] = \mu + \frac{\mathrm{Cov}(Y,\bar{Y})}{\mathrm{Var}(\bar{Y})}(\bar{Y}-\bar{\mu}) = \mu + \frac{\frac{1+2\rho}{3}\mathbf{1}}{\frac{1+2\rho}{3}}(\bar{Y}-\bar{\mu}) = \mu + \mathbf{1}(\bar{Y}-\bar{\mu}).$$

The conditional covariance is

$$\mathrm{Var}(Y \mid \bar{Y}) = \Sigma - \frac{\mathrm{Cov}(Y,\bar{Y})\,\mathrm{Cov}(Y,\bar{Y})^\top}{\mathrm{Var}(\bar{Y})}.$$

Since $\mathrm{Cov}(Y,\bar{Y}) = \frac{1+2\rho}{3}\mathbf{1}$, we have

$$\frac{\mathrm{Cov}(Y,\bar{Y})\,\mathrm{Cov}(Y,\bar{Y})^\top}{\mathrm{Var}(\bar{Y})} = \frac{\left(\frac{1+2\rho}{3}\right)^2\mathbf{1}\mathbf{1}^\top}{\frac{1+2\rho}{3}} = \frac{1+2\rho}{3}\mathbf{1}\mathbf{1}^\top.$$

Therefore

$$\mathrm{Var}(Y \mid \bar{Y}) = (1-\rho)I_3 + \rho\mathbf{1}\mathbf{1}^\top - \frac{1+2\rho}{3}\mathbf{1}\mathbf{1}^\top = (1-\rho)I_3 + \left(\rho - \frac{1+2\rho}{3}\right)\mathbf{1}\mathbf{1}^\top.$$

Now

$$\rho - \frac{1+2\rho}{3} = \frac{3\rho-1-2\rho}{3} = \frac{\rho-1}{3} = -\frac{1-\rho}{3}.$$

Hence

$$\mathrm{Var}(Y \mid \bar{Y}) = (1-\rho)I_3 - \frac{1-\rho}{3}\mathbf{1}\mathbf{1}^\top = (1-\rho)\left(I_3 - \frac{1}{3}\mathbf{1}\mathbf{1}^\top\right).$$

So

$$Y \mid \bar{Y} \sim N_3\Big(\mu + \mathbf{1}(\bar{Y}-\bar{\mu}),\ (1-\rho)\left(I_3 - \frac{1}{3}\mathbf{1}\mathbf{1}^\top\right)\Big).$$

Since $W = Y + q$ with q fixed, we simply add q to the conditional mean:

$$\boxed{W \mid \bar{Y} \sim N_3\Big(\mu + q + \mathbf{1}(\bar{Y}-\bar{\mu}),\ (1-\rho)\left(I_3 - \frac{1}{3}\mathbf{1}\mathbf{1}^\top\right)\Big).}$$

Note that $\bar{\mu} = (\mu_1+\mu_2+\mu_3)/3$ and $\bar{Y}$ is the observed average.

Problem 2

Let $X_i \overset{\text{ind}}{\sim} \text{Gamma}(\alpha_i, 1)$ *for* $i = 1, \ldots, p$. *Define*

$$Y_p = X_1 + \cdots + X_p, \qquad Y_i = X_i / Y_p \quad \text{for} \quad i = 1, \ldots, p-1.$$

Part (a)

Using the transformation technique, find the joint pdf of $Y = [Y_1, \ldots, Y_p]^\top$.

Theory: The transformation from X to Y is one-to-one. The joint density of Y is obtained by multiplying the joint density of X by the absolute value of the Jacobian determinant.

Solution: The joint density of $X_1, \ldots, X_p$ is

$$f_X(x_1, \ldots, x_p) = \prod_{i=1}^{p} \frac{1}{\Gamma(\alpha_i)} x_i^{\alpha_i - 1} e^{-x_i}, \quad x_i > 0.$$

Define the transformation:

$$y_i = \frac{x_i}{s} \quad (i = 1, \ldots, p-1), \qquad y_p = s = \sum_{i=1}^{p} x_i.$$

The inverse transformation is

$$x_i = y_i \, y_p \quad (i = 1, \ldots, p-1), \qquad x_p = y_p \left(1 - \sum_{i=1}^{p-1} y_i \right).$$

The Jacobian matrix of the inverse transformation is

$$J = \frac{\partial(x_1, \ldots, x_p)}{\partial(y_1, \ldots, y_{p-1}, y_p)} = \begin{pmatrix} y_p I_{p-1} & \mathbf{y} \\ -y_p \mathbf{1}^\top & 1 - \sum_{i=1}^{p-1} y_i \end{pmatrix},$$

where $\mathbf{y} = (y_1, \ldots, y_{p-1})^\top$ and $\mathbf{1}$ is a vector of ones.

Using the block determinant formula

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(A) \, \det(D - CA^{-1}B),$$

with $A = y_p I_{p-1}$, $B = \mathbf{y}$, $C = -y_p \mathbf{1}^\top$, $D = 1 - \sum y_i$, we get

$$D - CA^{-1}B = 1 - \sum_{i=1}^{p-1} y_i + \mathbf{1}^\top \mathbf{y} = 1.$$

Thus $\det(J) = \det(y_p I_{p-1}) \cdot 1 = y_p^{p-1}$. Since $y_p > 0$, the absolute value is y_p^{p-1}.

Now substitute into the density:

$$\begin{aligned} f_Y(y_1, \ldots, y_{p-1}, y_p) &= f_X(x_1, \ldots, x_p) \, |\det J| \\ &= \prod_{i=1}^{p} \frac{1}{\Gamma(\alpha_i)} \left(\prod_{i=1}^{p-1} (y_i y_p)^{\alpha_i - 1} \right) \left(y_p (1 - \sum y_i) \right)^{\alpha_p - 1} e^{-\sum x_i} \cdot y_p^{p-1}. \end{aligned}$$

But $\sum x_i = \sum_{i=1}^{p-1} y_i y_p + y_p(1-\sum y_i) = y_p$. Therefore $e^{-\sum x_i} = e^{-y_p}$.

Now collect powers of y_p:

$$\prod_{i=1}^{p-1} y_p^{\alpha_i - 1} \cdot y_p^{\alpha_p - 1} \cdot y_p^{p-1} = y_p^{\sum_{i=1}^{p-1}(\alpha_i - 1)+(\alpha_p-1)+(p-1)} = y_p^{(\sum_{i=1}^{p}\alpha_i)-1}.$$

The y_i part becomes

$$\prod_{i=1}^{p-1} y_i^{\alpha_i - 1} \cdot (1-\sum_{i=1}^{p-1} y_i)^{\alpha_p - 1}.$$

Thus the joint density is

$$\boxed{f_{Y_1,\ldots,Y_{p-1},Y_p}(y_1,\ldots,y_{p-1},y_p) = \frac{1}{\prod_{i=1}^{p}\Gamma(\alpha_i)} \left(\prod_{i=1}^{p-1} y_i^{\alpha_i-1}\right)\left(1-\sum_{i=1}^{p-1} y_i\right)^{\alpha_p-1} y_p^{\sum_{i=1}^{p}\alpha_i - 1} e^{-y_p},}$$

for $y_i > 0$ $(i = 1,\ldots,p-1)$, $\sum_{i=1}^{p-1} y_i < 1$, and $y_p > 0$.

Part (b)

Hence, find the joint pdf of $[Y_1,\ldots,Y_{p-1}]^\top$ and the marginal distribution of Y_p.

Theory: To obtain the marginal of a subset, integrate the joint density over the other variables. The integral over the simplex of a Dirichlet kernel yields the Dirichlet normalising constant.

Solution: First, integrate out y_p from the joint density obtained in part (a):

$$f_{Y_1,\ldots,Y_{p-1}}(y_1,\ldots,y_{p-1}) = \int_0^\infty f_{Y_1,\ldots,Y_{p-1},Y_p}(y_1,\ldots,y_{p-1},y_p)\, dy_p.$$

The y_p-dependent part is

$$\int_0^\infty y_p^{\sum \alpha_i - 1} e^{-y_p} dy_p = \Gamma\left(\sum_{i=1}^{p}\alpha_i\right).$$

Thus

$$\boxed{f_{Y_1,\ldots,Y_{p-1}}(y_1,\ldots,y_{p-1}) = \frac{\Gamma(\sum_{i=1}^{p}\alpha_i)}{\prod_{i=1}^{p}\Gamma(\alpha_i)} \left(\prod_{i=1}^{p-1} y_i^{\alpha_i-1}\right)\left(1-\sum_{i=1}^{p-1} y_i\right)^{\alpha_p-1},}$$

which is the Dirichlet density with parameters $(\alpha_1,\ldots,\alpha_p)$.

Next, to find the marginal distribution of Y_p, integrate out $y_1,\ldots,y_{p-1}$ from the joint density:

$$f_{Y_p}(y_p) = \int_{\mathcal{S}} f_{Y_1,\ldots,Y_{p-1},Y_p}(\cdots)\, dy_1 \cdots dy_{p-1},$$

where $\mathcal{S} = \{y_i > 0,\ \sum_{i=1}^{p-1} y_i < 1\}$. The integral over the simplex of the Dirichlet kernel is

$$\int_{\mathcal{S}} \prod_{i=1}^{p-1} y_i^{\alpha_i-1}(1-\sum y_i)^{\alpha_p-1} dy_1\cdots dy_{p-1} = \frac{\prod_{i=1}^{p}\Gamma(\alpha_i)}{\Gamma(\sum_{i=1}^{p}\alpha_i)}.$$

Therefore,

$$f_{Y_p}(y_p) = \frac{1}{\prod_{i=1}^p \Gamma(\alpha_i)} \cdot \frac{\prod_{i=1}^p \Gamma(\alpha_i)}{\Gamma(\sum \alpha_i)} \cdot y_p^{\sum \alpha_i - 1} e^{-y_p} = \frac{1}{\Gamma(\sum_{i=1}^p \alpha_i)} y_p^{\sum \alpha_i - 1} e^{-y_p}.$$

Hence

$$\boxed{Y_p \sim \text{Gamma}\left(\sum_{i=1}^p \alpha_i,\ 1\right).}$$

Moreover, the joint density factorises as the product of the Dirichlet density for $(Y_1, \ldots, Y_{p-1})$ and the Gamma density for Y_p. Therefore, $(Y_1, \ldots, Y_{p-1})$ and Y_p are independent.

Final answers:

- Joint pdf of $(Y_1, \ldots, Y_p)$ is as in the boxed expression above.
- $(Y_1, \ldots, Y_{p-1})$ has Dirichlet$(\alpha_1, \ldots, \alpha_p)$ distribution.
- Y_p has Gamma$(\sum \alpha_i, 1)$ distribution, independent of $(Y_1, \ldots, Y_{p-1})$.